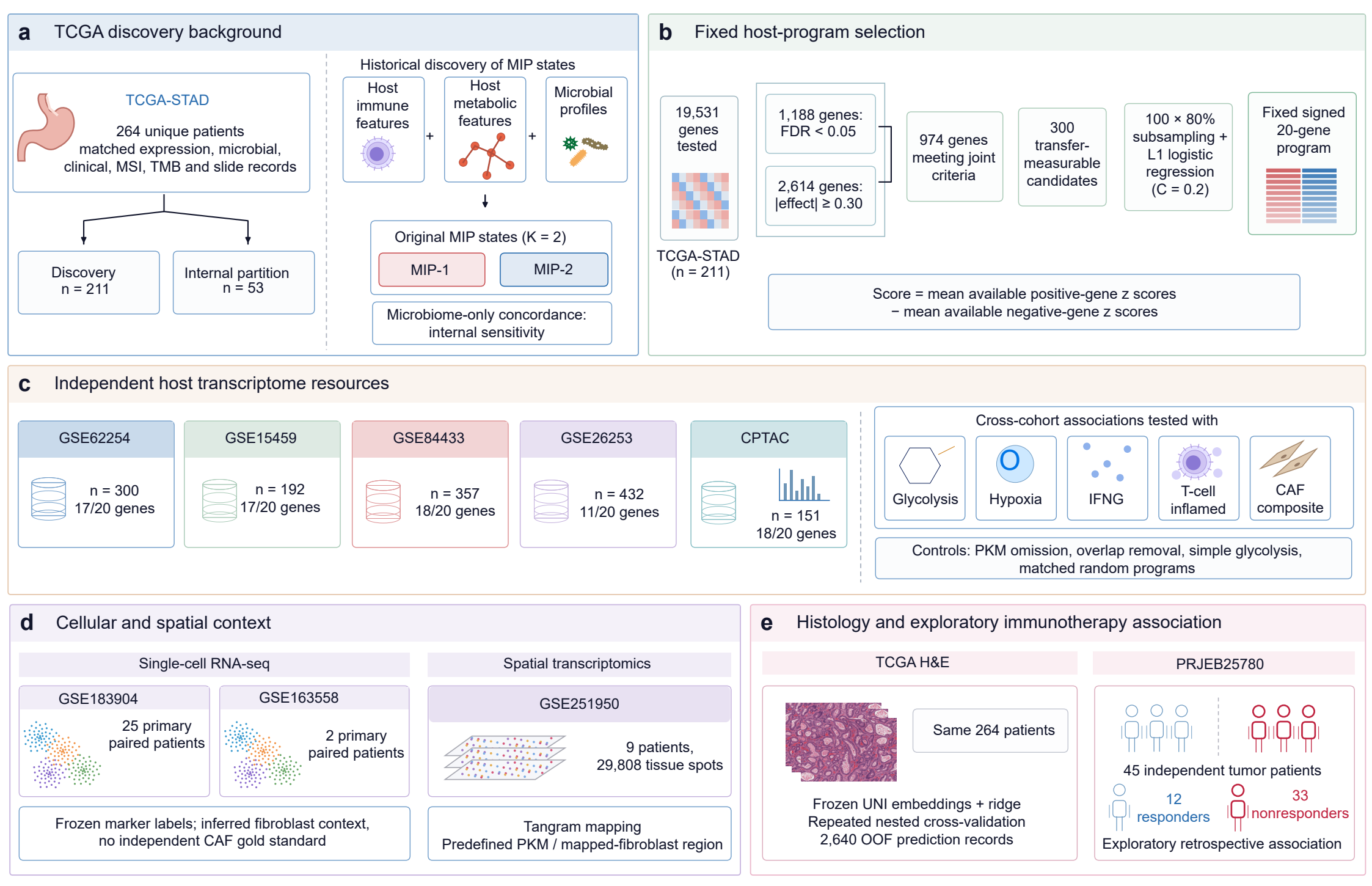

a
TCGA discovery background
TCGA-STAD
264 unique patients
matched expression, microbial,
clinical, MSI, TMB and slide records
Discovery
n = 211
Internal partition
n = 53
Historical discovery of MIP states
Host immune features
Host metabolic features
Microbial profiles
Original MIP states (K = 2)
MIP-1
MIP-2
Microbiome-only concordance:
internal sensitivity
b
Fixed host-program selection
19,531 genes tested
TCGA-STAD
(n = 211)
1,188 genes: FDR < 0.05
2,614 genes: |effect| ≥ 0.30
974 genes meeting joint criteria
300 transfer-measurable candidates
100 × 80% subsampling + L1 logistic regression (C = 0.2)
Fixed signed 20-gene program
Score = mean available positive-gene z scores
− mean available negative-gene z scores
c
Independent host transcriptome resources
GSE62254
n = 300
17/20 genes
GSE15459
n = 192
17/20 genes
GSE84433
n = 357
18/20 genes
GSE26253
n = 432
11/20 genes
CPTAC
n = 151
18/20 genes
Cross-cohort associations tested with
Glycolysis
Hypoxia
IFNG
T-cell inflamed
CAF composite
Controls: PKM omission, overlap removal, simple glycolysis,
matched random programs
d
Cellular and spatial context
Single-cell RNA-seq
GSE183904
25 primary paired patients
GSE163558
2 primary paired patients
Frozen marker labels; inferred fibroblast context,
no independent CAF gold standard
Spatial transcriptomics
GSE251950
9 patients,
29,808 tissue spots
Tangram mapping
Predefined PKM / mapped-fibroblast region
e
Histology and exploratory immunotherapy association
TCGA H&E
Same 264 patients
Frozen UNI embeddings + ridge
Repeated nested cross-validation
2,640 OOF prediction records
PRJEB25780
45 independent tumor patients
12 responders
33 nonresponders
Exploratory retrospective association